PHASE 3 RESEARCH STAY REPORT

PROGRAMA JOVES I CIÈNCIA - FUNDACIÓ CATALUNYA LA PEDRERA

and

Osservatorio Astronomico di Brera - Istituto Nazionale di Astrofisica (OAB-INAF)

Multi-Messenger Cosmology: Constraining $\Omega_m - \Omega_\Lambda$ Parameters with the $E_{\text{peak}} - E_{\text{iso}}$ Gamma-Ray Burst Relation and Simulated Gravitational Wave Events Detectable by Einstein Telescope

Author

Joan Alcaide-Núñez

Supervisors:

Dr. Giancarlo Ghirlanda

Dr. Om Sharan Salafia

October 2, 2025

1 Introduction

Since I was a kid, I have been fascinated by the most energetic, fast, big, and exotic phenomena that I knew of. First, planes, then rockets and astronauts, and now astrophysics. Cosmology is the study of the biggest structure: the Universe, its origin, evolution, and fate. While Gamma-Ray Bursts (GRBs) are the most energetic events that we know of, outshining the rest of the Universe when they appear. Combined they pose the research topic that interests me the most and that I had the opportunity to explore in this research stay.

During five weeks, I learnt from experts about how GRBs can help us calculate constraints on cosmological parameters, particularly the density parameters Ω_m and Ω_Λ. I had the chance to analyze data from the literature, enabling me to leverage my mathematical, especially statistical, and coding skills. Additionally, the project stretched beyond as we included simulated gravitational wave (GW) events to decrease the uncertainty on the measurements of the parameters. Multi-messenger is a flourishing field in astrophysics -the first event detected with both GW and EM signals was in 2017- and new gravitational observatories in the 2030s, like Einstein Telescope, are expected to increase our chances of detecting more of these events. Thus, it was very interesting to learn about the possibilities of this field from the hand of sophisticated simulations.

The atmosphere during the stay was very motivating, and having multiple people to rely on and that could answer my basic questions was good. Additionally, a former senior member of the group, Gabriele Ghisellini, gave me a 1-on-1 lecture each week about the foundations of relativistic plasma and radiative processes. I learnt a lot from these slides, helping me broaden my understanding of the physical nature of Gamma-Ray Bursts beyond their implications in Cosmology. Overall, the research stay was a very positive experience, in which I had a great time and I am looking forward to further exploring this topic.

2 Technical Details

Center	Osservatorio Astronomico di Brera - Istituto Nazionale di Astrofisica (OAB-INAF)
Location	Via Emilio Bianchi 46, 23807 Merate (Lc) – ITALY
Research Lines	• Galaxies and Cosmology • Star, Stellar Populations and the Interstellar Medium • Relativistic Astrophysics and Particles • Advanced Technologies & Instrumentation
Research Group	High Energy Astrophysics Group (HEA-Merate)
Supervisors	Dr. Giancarlo Ghirlanda, Dr. Om Sharan Salafia
Title	Multi-Messenger Cosmology
Timeframe	June 16 to July 16 2025

3 Aim of the Project

The initial aim of the project was to estimate cosmological parameters using the properties of GRBs. A variety of relations between GRB properties have been proposed and tested since the 2000s (e.g., [1]). One that stands out is the correlation between the isotropic equivalent energy $E_{\rm iso}$ and the peak energy $E_{\rm peak}$. Key to GRBs is that they are transient events, meaning they are one-time bursts exhibiting a light curve. $E_{\rm peak}$ is the maximum of this flux–time light curve. Additionally, the light curves of GRBs also show a break time (the time when the flux slope becomes steeper), hinting at their jet structure. The isotropic energy $E_{\rm iso}$ is the isotropic equivalent (i.e., if the GRBs had no jet structure but instead emitted in all directions), derived from the fluence F "corrected" for the volume and time travel:

$$E_{\rm iso} = \frac{4\pi d_L^2 F}{1+z},$$

where d_L is the luminosity distance. The $E_{\rm peak}$–$E_{\rm iso}$ correlation states that these properties are proportional, following a power law:

$$E_{\rm peak} = k \times E_{\rm iso}^a.$$

However, when working with the logarithms of $E_{\rm peak}$ and $E_{\rm iso}$, this becomes a linear equation:

$$E_{\rm peak} = a \times E_{\rm iso} + b,$$

where $b = \log_{10} k$. In order to estimate the cosmological parameters, I use the fact that $E_{\rm iso}$ depends on a specific cosmology through $d_L(z; H_0, \Omega_m, \Omega_\Lambda)$, thus making this relation cosmology-dependent. To find the actual values of the parameters, I fit the correlation for each possible combination of the $(\Omega_M, \Omega_\Lambda)$ plane and calculate its $\chi^2_{\rm reduced}$ value, which indicates the goodness of fit. By doing so, I was able to constrain the "best" cosmology, where the χ^2 value is minimized.

To apply this method, I used a luminosity distance function following the Friedmann equation and a ΛCDM Universe (using the `astropy` library):

$$d_L(z) = (1+z)\frac{c}{H_0}\int^z \frac{dz'}{\sqrt{\Omega_m(1+z')^3 + \Omega_\Lambda}},$$

where it is assumed that $\Omega_r = \Omega_k = 0.0$ (a flat Universe, neglecting radiation) and $H_0 = 70.0$ km s^{-1}Mpc^{-1}, in order to fit the $(\Omega_m, \Omega_\Lambda)$ parameter space. For the chi-squared test (χ^2), I applied the following equation:

$$\chi^2 = \sum_i \frac{\left[E_{{\rm peak},i} - \left(a \times E_{{\rm iso},i} + b\right)\right]^2}{\sigma_{E_{{\rm peak},i}} + \sigma_{\rm extra}}.$$

Here, $\sigma_{\rm extra}$ stands for systematic error and is added to account for when the relation's intrinsic scatter is greater than the observed errors. For our data, the table extracted from [2], this value was $\sigma_{\rm extra} = 0.114$.

With these tools, we may construct a $\chi^2(\Omega_m, \Omega_\Lambda)$ surface. Its minimum and contours indicate the estimated values for the computed cosmological parameters. I obtain: $\Omega_m = 0.121$, $\Omega_\Lambda = 0.974$, with $\chi^2 = 74.948$ and $\sigma_{\rm extra} = 0.114$.

Redshift z and luminosity distance d_L for GW signals can be precisely computed when there are electromagnetic counterparts, providing an independent method to measure cosmological parameters, as the combination of both signals offers additional information about the source's distance. This means one can determine the redshift and distance to a merger without assuming a specific cosmology, making these

events "standard sirens." Our goal was to fit the z and $d_L(z; H_0, \Omega_m, \Omega_\Lambda)$ in a Hubble diagram (assuming $H_0 = 70.0\ \mathrm{kms^{-1}Mpc^{-1}}$ as before) to obtain independent constraints for $(\Omega_m, \Omega_\Lambda)$. For the GW data I used the reduced χ^2 test, which uses eq. 3, adding the degrees of freedom (dof), which is the difference between number of data points and free parameters, to the denominator.

Since only one multi-messenger event has been observed so far [3], I used simulated data of a population of binary neutron star (BNS) mergers from [4], obtaining the results $\Omega_m = 0.340$, $\Omega_\Lambda = 0.919$ (Fig. 3).

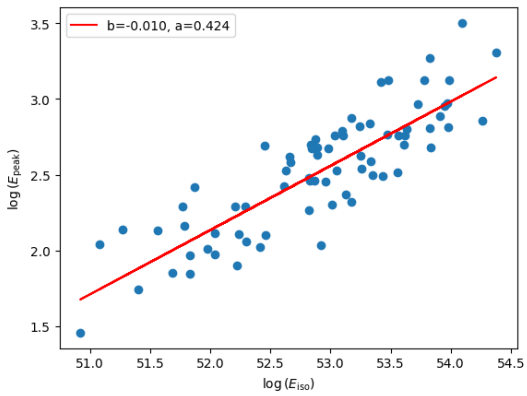

Figure 1: E_{peak}–E_{iso} correlation assuming a standard flat-Universe cosmology of $h = \Omega_\Lambda = 0.7$, and neglecting the density contribution of radiation. This figure was computed applying the barycenter correction and without adding any extra scatter. This was later implemented when iterating through the $(\Omega_m, \Omega_\Lambda)$ plane.

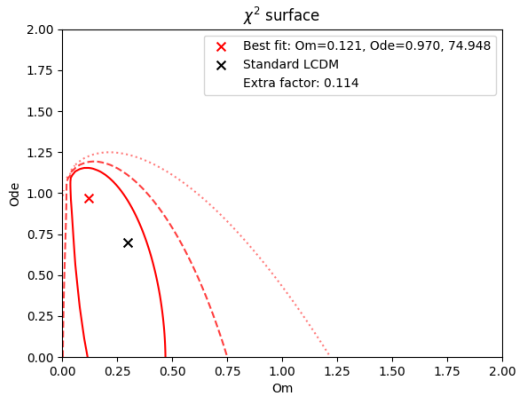

Figure 2: χ^2 surface in the $(\Omega_m, \Omega_\Lambda)$ plane showing the best-fit point and 68%, 90%, and 99% levels in red with the cross standing for the best fit. The dark cross represents the standard cosmology parameters.

Finally, I combine these two χ^2 surfaces—the one computed using the E_{iso}–E_{peak} relation for GRBs and the other using the Hubble diagram of BNS mergers by summing both matrices, obtaining the following results: $\Omega_m = 0.292$, $\Omega_\Lambda = 0.848$, with $\chi^2_{\mathrm{min}} = 76.007$. These results are consistent with the standard cosmology values. Nevertheless, although using simulated data, this shows the success of this method in computing the density parameters.

Future work includes using short GRB data associated with the BNS simulation instead of using separate long GRB data for the cosmological constraints. Moreover, one could explore the effect of sensitivity and observation rate of events with new observatories, such as the Einstein Telescope (ET), to forecast what precision levels can we achieve with specific number of detections.

4 Conducted Activities

Before the stay and at the beginning of it, my tasks mainly consisted of reading papers and taking notes, aided by the review provided by my supervisors. Later on, I started working with data, first searching, retrieving, and cleaning it, and then analyzing it using Python and multiple libraries to compute fits and find results. Even after prior reading, I still spent some time reaching for papers and documentation. A couple days, the institute hosted talks by external researchers which I also gladly attended.

Additionally, I attended the weekly journal clubs within the HEA group, in which I learned a lot about the high energy astrophysics field in a broader sense, beyond what my project could cover. I also benefited from conversations with Gabrielle Ghisellini, a retired member of the group, who taught me many topics from high energy astrophysics in a general way, drawing from his book.

Aside from the scientific project, I also participated in other activities at the observatory during my stay. On the 19th of June, the observatory inaugurated an LGBTQIA+ bench to coincide with Pride Month. During

the event, held in front of one of the domes, we listened to colleagues and reflected on the history of the movement and Pride in Merate.[1] I also attended a presentation by high school students from the region who had spent one week at the engineering section of the observatory and showcased their learnings to the staff.

Since the observatory still holds two telescope domes, one of which is still operational (the other being 100 years old), it hosts night events to observe the sky. I participated in one of these events, during which we had a Q&A about the observatory and the work of an astronomer, and then observed multiple objects with the Ruths telescope. This is the largest telescope in Italy, with a diameter of 137 cm. On another night, we also listened to a performance by two local musicians—a trumpeter and a saxophonist—in the Zeiss dome with the older telescope.[2] These events were quite remarkable.

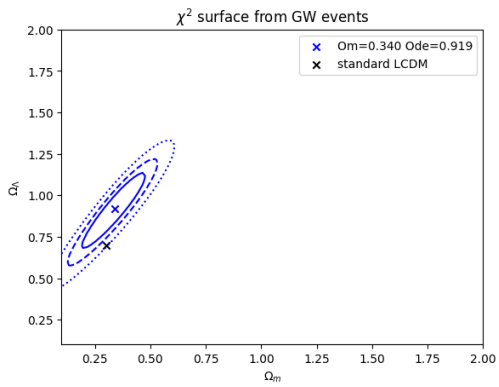

Figure 3: Same as fig 2, these figure shows the contours for the 68%, 90%, and 99% levels with a cross in its center showing the best fit (minimum χ^2). Black cross shows the standard cosmology ($\Omega_m = 0.3, \Omega_\Lambda = 0.7$).

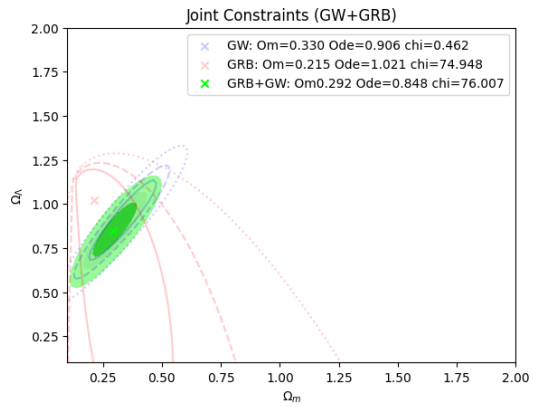

Figure 4: Filled with green, the 68%, 90%, and 99% contours for the joint constraints (GRB+GW), with a best fit of ($\Omega_m = 0.292, \Omega_\Lambda = 0.848$). In the background, the GRB and GW constraints from figures 2 and 3 are shown in red and blue respectively.

5 Reflection

All in all, I have had a very good impression of the field and the workflow, which is ultimately the goal of the *Joves i Ciència* Program and this research stay. From both the research activities and the outreach activities, I take great memories and new knowledge. Regarding the accommodation and atmosphere at the office and the guest house, they were very good, and I am happy to have gotten to know many people, not only from the field I was working on but also from other labs or, inclusively, researchers from other universities coming to the guest house. I think that conducting this project has enabled me to learn many mathematical tools which will serve me well during my physics studies and beyond. My ntention is to continue with this project beyond the duration of the stay. First, by continuing with additional steps outlined above and later on into the field.

Acknowledgments

First, I want to express my gratitude to the *Osservatorio Astronomico di Brera, Sede di Merate* for hosting me and providing a place to stay at the guest house as well as to Dr. Giancarlo Ghirlanda and Dr. Om Sharan Salafia for organizing the stay and allowing me to come to the group. I also want to extend my thanks to the other members of the group for the kind help and patience and the conversations on the chalkboards. Finally, I am grateful to the *Joves i Ciència* Program and my family for supporting me on this research stay abroad.

[1] https://brera.inaf.it/19-giugno-2025-una-panchina-lgbtqia-a-merate/
[2] https://brera.inaf.it/generazione-stellare-lorenzo-simoni-al-sax-contralto-e-iacopo-teolis-alla-tromba-merate-suonimobili/

References

1. Ghirlanda, G., Ghisellini, G. & Firmani, C. Gamma-ray bursts as standard candles to constrain the cosmological parameters. *New Journal of Physics* **8,** 123–123. ISSN: 1367-2630. http://dx.doi.org/10.1088/1367-2630/8/7/123 (July 2006).

2. Ghirlanda, G., Nava, L., Ghisellini, G., Firmani, C. & Cabrera, J. The E peak–E iso plane of long gamma-ray bursts and selection effects. *Monthly Notices of the Royal Astronomical Society* **387,** 319–330 (2008).

3. Abbott, B. P. *et al.* GW170817: Observation of Gravitational Waves from a Binary Neutron Star Inspiral. *Phys. Rev. Lett.* **119,** 161101. https://link.aps.org/doi/10.1103/PhysRevLett.119.161101 (16 Oct. 2017).

4. Colombo, A. *et al.* Multi-messenger observations in the Einstein Telescope era: binary neutron star and black hole-neutron star mergers. *arXiv preprint arXiv:2503.00116* (2025).